



HANDS-ON SESSION

Understand the Impact of Triggers During SAP S/4HANA Conversion

Using the Trigger Simulation Feature of Software Update Manager (SUM)

Session Level	Intermediate / Advanced
Estimated Duration	20 Minutes
Version	1.0 2026-08-26
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Prerequisites

- A completed SAP S/4HANA conversion with PUTTB_SHD.ZIP available
- SUM tool running and positioned at the 'Activate Triggers' phase
- Access to the SUM web interface (URL and credentials provided in the exercise sheet)
- Authorizations to run: BC_GLOBAL_SCUST_CREA, BC_GLOBAL_SBOOK_CREA, BC_GLOBAL_SBOOK_DISP
- Basic familiarity with SUM phases and SAP S/4HANA conversion lifecycle

Time Saver — Starting Point

To save time, the first few setup steps have been pre-completed for this session. The SUM tool is already loaded with PUTTB_SHD.ZIP and is currently paused at the 'Activate Triggers' phase. You will start from Exercise 1 — launching SUM.

Table of Contents

Overview & Learning Objectives	3
Reference: Trigger Modes at a Glance	3
Exercise 1: Launch SUM and Activate Triggers	4
Exercise 2: Explore the Trigger Control Center	5
Exercise 3: Observe FREEZE_LOG Behavior — Baseline	6
Exercise 4: Change Trigger Mode to FREEZE_LOG_ERROR	7
Exercise 5: Observe Business Impact — Blocked Transaction	8
Exercise 6: Switch to FREEZE_OFF and Compare	10
Summary & Key Takeaways	11

Overview & Learning Objectives

During an SAP S/4HANA conversion, the Software Update Manager (SUM) places database triggers on all tables to record the changes to data. These triggers define how the system responds to DML operations (INSERT, UPDATE, DELETE) on classified tables — particularly during the phases where those tables must be protected to ensure data consistency.

In a production cutover, the behavior of these triggers directly impacts running business processes. This session gives you hands-on experience on how triggers can impact your business process while the SAP S/4HANA conversion is running on the production system.

By the end of this session, you will be able to:

- Launch the SUM tool and advance the trigger activation step
- Navigate the Trigger Control Center and filter table-specific trigger status
- Explain the differences between FREEZE_LOG, FREEZE_ERROR, and FREEZE_LOG_ERROR
- Demonstrate the business impact of FREEZE triggers using SFLIGHT transactions
- Identify affected transactions and records using UPGILLACCESS and SBOOK display
- Deactivate triggers (FREEZE_OFF) and confirm normal transaction behavior resumes

Reference: Trigger Modes at a Glance

The following table summarizes the three trigger modes you will work with in this session. Keep this as a reference throughout the exercises.

Trigger Mode	DML Impact	Logging
FREEZE_LOG	DML statements allowed (not blocked)	Access logged in UPGILLACCESS
FREEZE_ERROR	DML statements blocked immediately	No logging
FREEZE_LOG_ERROR	DML statements logged and blocked immediately	Access logged (first instance only)

About the UPGILLACCESS Table

This table captures all DML accesses on FREEZE-mode tables. It is not registered in the ABAP Data Dictionary (inactive in SE11) — you must use DB-level tools (e.g., DB02, SE16 with a direct DB query, or your DBMS console) to query it. It is used to audit which tables were accessed during the freeze window.

EXERCISE 1

Launch SUM and Activate Triggers

Objective: Access the SUM web interface, log in, review the current trigger phase, and advance the process to activate triggers on all classified tables.

Background

The SUM tool pauses before each major trigger step to allow the BASIS team to review the current state. At this point the triggers have been created on the tables (based on the PUTTB_SHD.ZIP classification) but are not yet active. Clicking Next will activate all triggers simultaneously.

Caution — Production Impact Awareness

In a real production cutover, activating triggers is an irreversible step in the conversion window. Ensure all pre-checks and stakeholder notifications are complete before proceeding. In this exercise environment it is safe to proceed.

Steps

SUM Tool URL

```
https://<IP from exercise  
sheet>:1129/lms1/sumabap/ABA/slui
```

1

Open a browser and navigate to the SUM Tool URL shown above.

The IP address is provided on your exercise sheet.

2

Enter the login credentials when prompted:

Username: abaadm

Password: Pa\$\$word2026



Your connection isn't private

Attackers might be trying to steal your information from **100.58.109.0** (for example, passwords, messages, or credit cards).

NET::ERR_CERT_AUTHORITY_INVALID

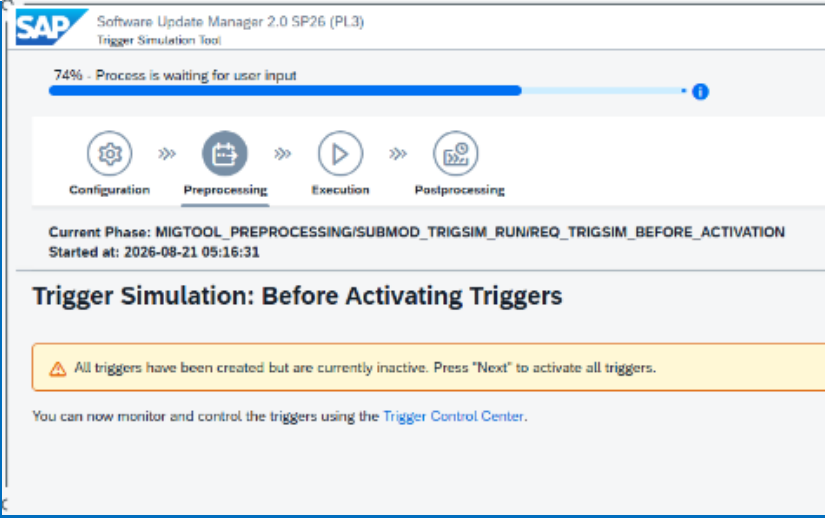
Hide advanced
Go back

This server couldn't prove that it's **100.58.109.0**; its security certificate is not trusted by your computer's operating system. This may be caused by a misconfiguration or an attacker intercepting your connection.

Continue to 100.58.109.0 (unsafe)

If a browser security warning appears (untrusted certificate), click 'Continue to site' or the equivalent option.

This is expected in the lab environment as a self-signed certificate is used.



The screenshot shows the SAP Software Update Manager 2.0 SP26 (PL3) Trigger Simulation Tool. A progress bar at the top indicates 74% completion, with the status 'Process is waiting for user input'. Below the progress bar, four icons represent the stages: Configuration, Preprocessing, Execution, and Postprocessing. The 'Preprocessing' stage is currently active. The current phase is 'MIGTOOL_PREPROCESSING/SUBMOD_TRIGSIM_RUN/REQ_TRIGSIM_BEFORE_ACTIVATION', which started at 2026-08-21 05:16:31. The main heading is 'Trigger Simulation: Before Activating Triggers'. A yellow warning box states: 'All triggers have been created but are currently inactive. Press "Next" to activate all triggers.' Below this, it says: 'You can now monitor and control the triggers using the Trigger Control Center.'

Review the current SUM phase displayed on screen.

The screen should show that triggers have been created but not yet activated. You are at the 'Activate Triggers' checkpoint.

5

Click the Next button to activate all triggers.

SUM will activate triggers on all classified tables simultaneously. Wait for the confirmation before proceeding to Exercise 2.

Expected Result

SUM confirms that triggers have been activated. The tool proceeds to the next phase. All tables classified in PUTTB_SHD.ZIP now have active FREEZE_LOG triggers.

EXERCISE 2

Explore the Trigger Control Center

Objective: Navigate to the Trigger Control Center within SUM utilities and verify the trigger status on table SBOOK.

Background

The Trigger Control Center (TCC) is the operational hub for managing active triggers during a conversion. It allows you to view the trigger mode for each table, filter by category or table name, and change modes individually or in bulk. In this exercise, you will locate the SBOOK table and confirm its initial status.

About Table SBOOK

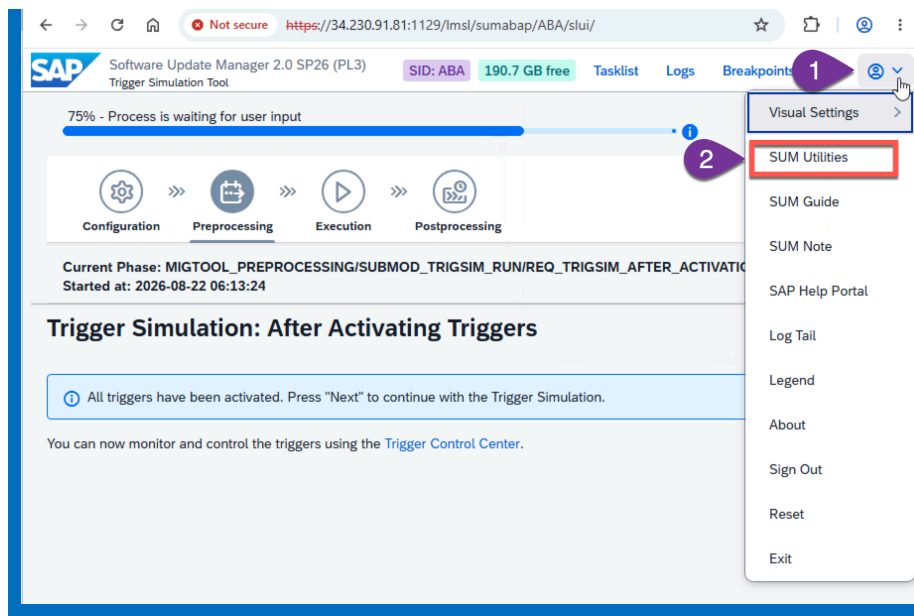
SBOOK is the flight booking table in the SFLIGHT demo application. It is a transactional table with frequent inserts and updates — making it an ideal table to demonstrate the business impact of FREEZE triggers in this exercise.

Navigation Path

Use the following menu path inside the SUM tool to reach the Trigger Control Center:

```
SUM Tool Main Menu  
→ SUM Utilities  
→ Process Control Center  
→ Trigger Control Center
```

Steps



In the SUM tool, navigate to:
SUM Utilities → Process
Control Center → Trigger
Control Center.

*Use the menu path shown
above.*

Trigger Control Center

Monitor and Control SUM Triggers

85604

All Triggers

85604

Active Triggers

0

Inactive Triggers

0

Deleted Triggers

Category: All Triggers
Action: Switch Off (UKR_OFF/FREEZE_OFF)
Apply
Active Inactive Deleted

Table Name	Trigger Name	Trigger Status	Trigger Request	Trigger Timestamp
Sort Ascending	1	UKR_LOG		2026-08-22 02:26:55
Sort Descending	2	UKR_LOG		2026-08-22 02:29:02
Filter SBOOK	0	UKR_LOG		2026-08-22 02:31:18

In the Trigger Control Center filter area, enter SBOOK in the Table Name field.

Apply the filter to narrow the list to the SBOOK table only.

3

Locate the entry for SBOOK and note the Trigger Type and Status columns.

Observe the current mode and whether the trigger is active.

4

Note the trigger category shown for SBOOK.

It should show a FREEZE category trigger.

Expected Result

SBOOK is displayed with Trigger Type FREEZE and current mode FREEZE_LOG. This means DML access is not yet blocked — it is only being recorded. The log is written to the UPGILLACCESS table, which is not visible in SE11.

UPGILLACCESS — Where to Check

Since UPGILLACCESS is inactive in the ABAP Data Dictionary, you cannot open it via SE16. Use transaction DB02 → Tables and Indexes, or consult your DBMS console (e.g., HANA Studio, DB13) to query the table directly at the database level.

EXERCISE 3

Observe FREEZE_LOG Behavior — Baseline

Objective: Run a flight booking transaction while SBOOK is in FREEZE_LOG mode and confirm that the transaction succeeds — and that access is logged.

Background

FREEZE_LOG mode is the initial, non-disruptive trigger state. DML operations on the table are allowed to complete normally, but each access is recorded in UPGILLACCESS. This is typically used during the analysis and rehearsal phase to understand which transactions touch a given table — before deciding to move to a more restrictive freeze mode.

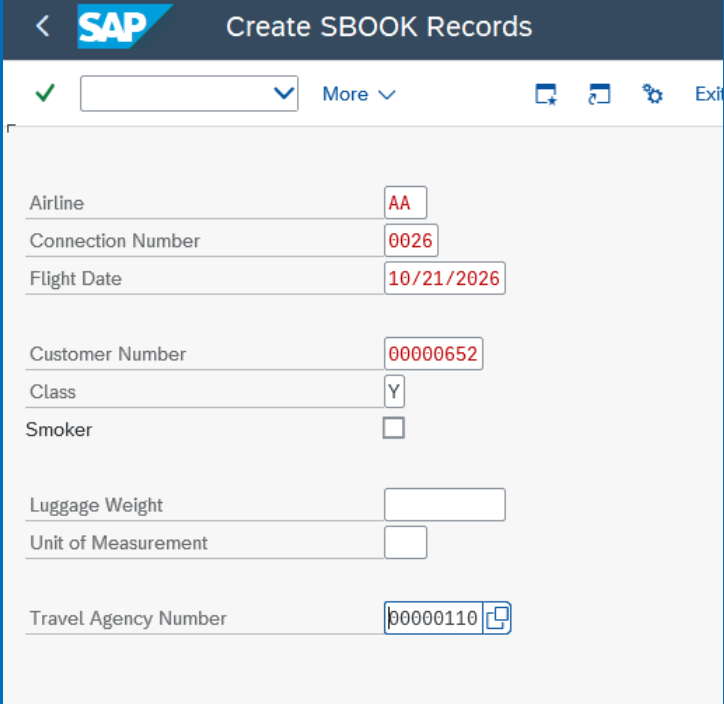
In this exercise you will create a flight booking using transaction BC_GLOBAL_SBOOK_CREA and confirm that the booking succeeds. This establishes your baseline before you change the trigger mode in Exercise 4.

Steps

1

In the SAP system (separate from the SUM browser), run transaction:

`BC_GLOBAL_SBOOK_CREA`



Enter valid values for the booking fields.

Use F4 help to select values for each field (airline, flight number, date, customer). Any valid combination works — the goal is to trigger a write to SBOOK.

SAP Create SBOOK Records

✓ [Dropdown] More ▾

Save (Ctrl+S)

Cancel (F12)

Airline

Connection Number 0026

Click Save (or select Save from the menu).

The booking should complete without error.

✓ Customer Adam Buchholm is booked on flight AA0026 on 10/21/2026 (booking number 00096901)

Note the booking number displayed in the confirmation message.

You will use this booking number in Exercise 5 to verify blocking behavior.

```
1> select * from UPGILLACCESS
2> go
  TABNAME      TIMESTAMP      UNCRITICAL
-----
SBOOK          20260822100505
```

Return to the Trigger Control Center and observe whether a new entry appears in UPGILLACCESS.

(Optional, if DB-level access is available.) Confirm the booking DML was logged even though the transaction succeeded.

Expected Result

The flight booking is created successfully — confirming that FREEZE_LOG mode does not block DML operations. The booking number is shown in the confirmation message. In the background, the access has been logged to UPGILLACCESS.

Tip — Real Production Use of FREEZE_LOG

In a real project, you would keep triggers in FREEZE_LOG mode for several hours, days, or up to a full week before the cutover — capturing representative access patterns. The UPGILLACCESS data then reveals exactly which tables are actively used, and by which transactions, during your planned freeze window.

EXERCISE 4

Change Trigger Mode to FREEZE_LOG_ERROR

Objective: Switch the SBOOK trigger from FREEZE_LOG to FREEZE_LOG_ERROR to simulate the production cutover trigger state.

Background

FREEZE_LOG_ERROR is the mode that most closely represents production cutover behavior. DML operations are blocked and an error is returned to the calling program. Additionally, the first DML access after the mode switch is logged in UPGILLACCESS. This is the mode you would use during the actual downtime window to protect table consistency.

Trigger Mode Reference

FREEZE_LOG → DML allowed, access logged

FREEZE_ERROR → DML blocked immediately, no logging

FREEZE_LOG_ERROR → DML blocked immediately, first access logged

Steps

Trigger Control Center
Monitor and Control SUM Triggers

27201 All Triggers 27201 Active Triggers 0 Inactive Triggers 0 Deleted Triggers

Category: Freeze Triggers Action: Switch On (FREEZE_LOG) Apply Active Inactive Deleted

Table Name	Trigger Name	Trigger Status	Trigger Request	Trigger Timestamp
Sort Ascending	/TF00055280	FREEZE_LOG		2026-08-22 01:23:42
Sort Descending				
Filter SBOOK				

In the Trigger Control Center, set the Category filter to 'Freeze Triggers'.

This narrows the list to only FREEZE-type triggers.

Trigger Control Center
Monitor and Control SUM Triggers

27201 All Triggers 27201 Active Triggers 0 Inactive Triggers 0 Deleted Triggers

Category: Freeze Triggers Action: Switch On (FREEZE_LOG) Apply

Table Name	Trigger Name	Trigger Status
SBOOK	/1CRR/TF00055280	FREEZE_LOG

In the Table Name filter, enter SBOOK and apply the filter.

The list should show a single row for SBOOK with its current mode.

Trigger Control Center

Monitor and Control SUM Triggers

27201

All Triggers

27201

Active Triggers

0

Inactive Triggers

0

Deleted Triggers

Category:

Freeze Triggers

Action:

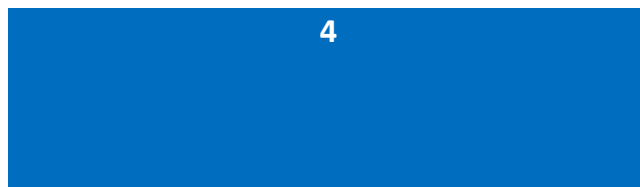
Switch On (FREEZE_LOG)

Apply

☒	Table Name	Trigger Name	Trigger Status
☒	SBOOK	/1CRR/T	FREEZE_LOG

Select the row for SBOOK by checking its checkbox.

Ensure only SBOOK is selected before proceeding.



Open the Action dropdown and select FREEZE_LOG_ERROR.

The action applies the new trigger mode to the selected table.



Click the Apply button.

Wait for confirmation that the trigger mode has been updated.

Category:

Freeze Triggers

Action:

Switch On (FREEZE_LOG_ERROR)

Apply

☐	Table Name	Trigger Name	Trigger Status	Trigger Request
☐	SBOOK	/1CRR/TF00055280	FREEZE_LOG	Switching On (FREEZE_LOG_ERROR) ...

Verify the Trigger Mode column for SBOOK now shows FREEZE_LOG_ERROR.

Refresh the list if necessary.

Expected Result

The SBOOK trigger mode is now FREEZE_LOG_ERROR. Any subsequent DML operation on SBOOK will be blocked and the first access will be logged. You are now ready to observe the business impact in Exercise 5.

Bulk Actions

The Action dropdown and Apply button support bulk operations — you can select multiple tables and change their trigger modes simultaneously. This is how you would manage hundreds of tables during a real production cutover window.

EXERCISE 5

Observe Business Impact — Blocked Transaction

Objective: Attempt to create a new flight booking after setting FREEZE_LOG_ERROR on SBOOK. Observe the error behavior, check the inbox, and verify no booking was created.

Background

This exercise simulates what a business user would experience if they attempted to perform a transaction on a FREEZE-protected table during the conversion downtime window. Even though the system appears to be running, certain tables are protected — and the results are immediate and visible.

This is the Production Reality

In a real cutover, users who have not been properly informed about the downtime window may attempt to book flights, process orders, or update records — and encounter exactly this behavior. The trigger simulation gives you the evidence to communicate the impact clearly to business stakeholders before go-live.

Steps — Create a Booking (Will Fail)

The screenshot shows the SAP 'Create SBOOK Records' transaction. The form is titled 'Create SBOOK Records' and has a green checkmark icon. The fields are as follows:

Field	Value
Airline	AA
Connection Number	0026
Flight Date	10/21/2026
Customer Number	00000652
Class	Y
Smoker	☐
Luggage Weight	
Unit of Measurement	
Travel Agency Number	00000110

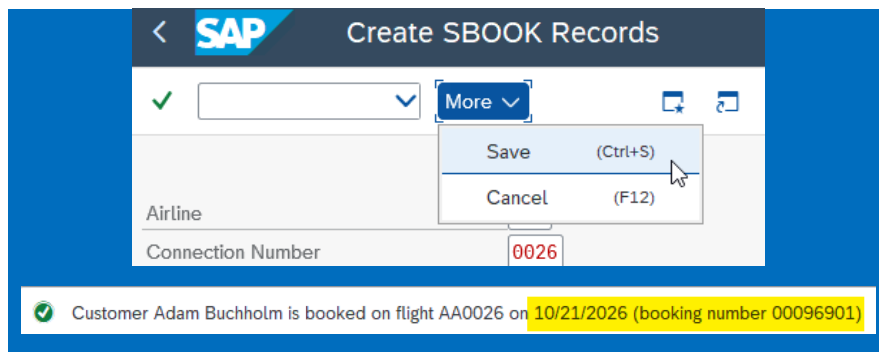
Run transaction BC_GLOBAL_SBOOK_CREA.

This is the same flight booking creation transaction used in Exercise 3.

2

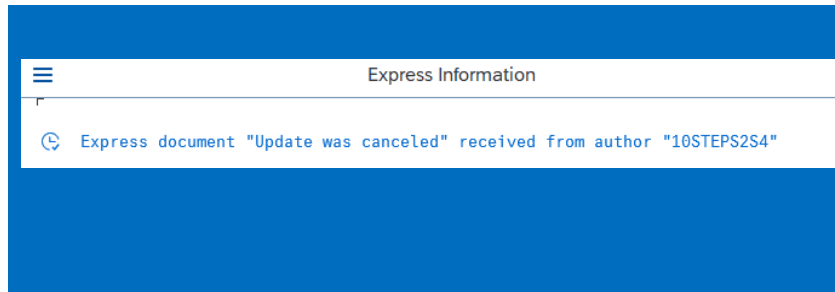
Enter valid booking values (use F4 to browse options).

Select a different customer or flight from the one you used in Exercise 3 so the comparison is clear.



Click Save.

You will first see a booking confirmation message — but observe what happens next.



Note the failure notification that appears immediately after the confirmation.

The system shows an error message indicating the booking was blocked by a trigger.



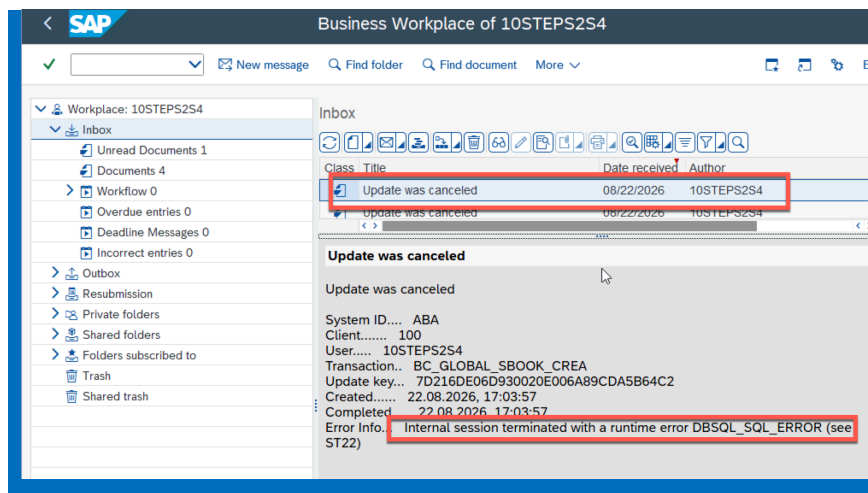
Note the booking number shown in the confirmation before the error.

Keep this number for the next step.

Expected Result

A booking confirmation appears briefly, immediately followed by a FREEZE-related error notification. The transaction does not complete successfully — the booking is not persisted in SBOOK.

Steps — Verify in Inbox and Display Transaction



Open your SAP Business Workplace inbox (or use the notification icon).

Navigate to the inbox to see the FREEZE error message that was routed there.

Category	Installation Errors
Runtime Errors	DBSQL_SQL_ERROR
Except.	CX_SY_OPEN_SQL_DB
Date and Time	08/22/2026 17:03:57
Short Text	
SQL error "SQL code: 200000" occurred while accessing table "SBOOK".	
What happened?	
Database error text: "SQL message: [ASE Error SQL200000][SAP][ASE ODBC Driver][Adaptive Server Enterprise]FREEZE-002: No changes allowed on table SBOOK due to maintenance. (LOGTAB: UPGILLACCESS)"	

Review the error message in the inbox.

The message will reference the FREEZE trigger that blocked the operation.

 SBOOK Record Display:

✓ Cancel

Airline

Booking number

 Display

Run transaction
BC_GLOBAL_SBOOK_DISP.

This is the flight booking display transaction.

 Booking 00096901 not found (table SBOOK)

Search for the booking number from Step 5.

Enter the booking number in the appropriate search field.

Expected Result

The booking display confirms the transaction did not go through — the booking record either does not exist or shows an incomplete/rolled-back state. The inbox contains the FREEZE error log entry with details of the blocked operation.

What This Means for Your Cutover Plan

Every table in FREEZE_LOG_ERROR mode will produce this behavior for any transaction that writes to it. Before activating FREEZE triggers in production, ensure:

- All users are locked out of the system (system-level lockout is the safest approach)
- Batch jobs and interfaces are stopped
- SLT and replication channels are paused
- Go / No-Go confirmation is received from all workstream leads

EXERCISE 6

Switch to FREEZE_OFF and Compare Behavior

Objective: Change the SBOOK trigger mode to FREEZE_OFF, create a new booking, and confirm that normal transaction behavior is restored.

Background

FREEZE_OFF deactivates the trigger for the selected table, restoring normal DML operations. In a real conversion, this would be done after the XCLA/DMO step completes for that table — the table is no longer in a protected state and business operations can resume. Observing the contrast between FREEZE_LOG_ERROR and FREEZE_OFF makes the impact of trigger modes concrete and demonstrable.

Steps — Deactivate the Trigger

1	Return to the Trigger Control Center (Exercise 2 navigation path). <i>SUM Utilities → Process Control Center → Trigger Control Center</i>
2	Filter by Category: Freeze Triggers and Table Name: SBOOK. <i>Confirm SBOOK is currently in FREEZE_LOG_ERROR mode.</i>
3	Select the SBOOK row and open the Action dropdown.
4	Select FREEZE_OFF from the action list and click Apply. <i>Wait for confirmation that the mode has been updated.</i>
5	Verify the Trigger Mode column for SBOOK now shows FREEZE_OFF.

Steps — Confirm Normal Behavior

6	Run transaction BC_GLOBAL_SBOOK_CREA again. <i>Enter valid booking values using F4.</i>
7	Click Save.

	Observe the transaction behavior carefully — compare to what happened in Exercise 5.
--	--

8	Note the booking confirmation and booking number. <i>No error should appear after the confirmation.</i>
---	--

9	Run BC_GLOBAL_SBOOK_DISP and search for the new booking number. <i>Confirm the booking record exists and is complete.</i>
---	--

Expected Result

The booking completes successfully with no error. The booking record is visible in BC_GLOBAL_SBOOK_DISP. Normal DML operations on SBOOK are fully restored. This confirms the before/after behavior of FREEZE trigger modes.

Explore Further

Try the same sequence (FREEZE_LOG → FREEZE_LOG_ERROR → FREEZE_OFF) with the customer creation transaction BC_GLOBAL_SCUST_CREA to observe the same pattern on a different table. Also explore bulk mode changes across multiple tables at once to simulate the scale of a real production cutover.

Summary & Key Takeaways

This session gave you hands-on experience with the full trigger lifecycle inside the SUM tool — from activation through mode changes to deactivation. You observed both the silent monitoring behavior of FREEZE_LOG and the disruptive blocking behavior of FREEZE_LOG_ERROR, using real SFLIGHT transactions to make the impact tangible.

Exercise	Key Takeaway
Exercise 1: Launch SUM & Activate Triggers	The SUM tool pauses before each trigger phase to allow review. Clicking Next activates all classified table triggers simultaneously.
Exercise 2: Trigger Control Center	The TCC is the operational interface for monitoring and managing trigger modes. UPGILLACCESS stores the access log but is only visible at the DB level.
Exercise 3: FREEZE_LOG Baseline	FREEZE_LOG allows all DML to proceed normally while silently logging access. Use it during rehearsal to identify active tables without disrupting users.
Exercise 4: Mode Change to FREEZE_LOG_ERROR	Trigger mode changes can be done individually or in bulk. FREEZE_LOG_ERROR is the closest representation of production cutover behavior.
Exercise 5: Business Impact Observation	FREEZE_LOG_ERROR blocks DML immediately. Users see an error after a brief confirmation. The inbox and SBOOK display confirm the transaction was not persisted.
Exercise 6: FREEZE_OFF — Normal Behavior Restored	Setting FREEZE_OFF removes the trigger protection. Transactions succeed as expected. This is the state after a table's conversion step completes in production.

Broader Context — Planning Your Cutover

Run the simulation well before your production cutover, document which transactions are blocked, and use that evidence to build your business-facing downtime impact assessment. The combination of SUMToolBox impact analysis (covered in the companion session) and trigger simulation gives you a complete, data-driven picture of conversion risk.

SAP Help — SLT Toolset & Trigger Simulation:

<https://help.sap.com/docs/SLTOOLSET/b0d775c8a9fd47b985a5e66ddba31d33/53d7780f8f6c478c9ed12ae995ff8733.html?locale=en-US>